# Journey Inside the Atom

Chapter 8 | Exploration, Grade 9 (NCERT) | Quick Revision Notes

## 8.1 Early Ideas about the Atom

- **Acharya Kanada** (India, *Vaisesika Sutras*): repeated division of matter ends at indivisible *parmanus*; combinations form dyads (2) and triads (3). **Leucippus and Democritus** (Greece) called them *atomos* (indivisible). These were **ideas, not experimental results**.
- **John Dalton (1808)**: first scientific atomic theory - all matter is made of indivisible atoms.
- Late 19th century: **radioactivity** (some elements emit radiation) showed atoms are made of smaller particles, so atoms are **divisible**. Atoms are neutral and do not have colour (colours in diagrams are only illustrative).

## 8.2 Evolution of Atomic Models

### *Thomson's model (plum pudding / watermelon model)*

- **J. J. Thomson (1897)** passed current through gas at very low pressure in a cathode ray tube: rays moved from cathode to anode. In electric and magnetic fields they behaved as **negatively charged particles much lighter than atoms** = **electrons** (first subatomic particle). Cathode rays were the same whatever the cathode or gas, so electrons are present in all atoms.
- Charge of electron = $-1.602 \times 10^{-19}$ C, taken as **-1** by convention.
- **Model:** atom = sphere of positive charge with electrons embedded in it (like seeds in a watermelon), so the atom is neutral overall.

### *Rutherford's model (nuclear / planetary model) - gold foil experiment*

- **Geiger and Marsden (1911, under Rutherford)** fired a narrow beam of **alpha particles** (positively charged, = helium nucleus with 2 protons + 2 neutrons) at a very thin gold foil. This is also called the **alpha-ray scattering experiment**.

| Observation | Conclusion |
|---|---|
| **Most alpha particles passed straight through** | Most of the atom is **empty space** |
| **Some were deflected through small/large angles** | Positive charge is concentrated in a tiny centre (**nucleus**); like charges repel |
| **A very few bounced back** | Nucleus is **dense and heavy** and holds all the positive charge and most of the mass |

- Electrons revolve around the nucleus like planets around the Sun (**planetary model**). Atom diameter about $10^{-10}$ m; nucleus about $10^{-15}$ m (about $10^{5}$ times smaller). Thomson's model failed to explain this result.
- **Limitation:** a revolving electron is accelerating, so it should lose energy, spiral inward and fall into the nucleus, making atoms unstable. But atoms are stable. Rutherford's model could not explain this.
- **Proton** (discovered by Rutherford): positive charge (+1), much heavier than electron. **Number of protons = number of electrons** in a neutral atom (He: 2p, 2e; Na: 11p, 11e).

### *Bohr's model (1913) - explains stability*

- Electrons move only in fixed circular paths called **orbits / shells / stationary states / energy levels**, not in between.
- While in a shell, an electron **does not lose energy** (stationary state - a postulate), so it does not fall into the nucleus.
- Shells are named **K, L, M, N...** (n = 1, 2, 3, 4...). K-shell is nearest the nucleus and has the **lowest energy**; energy increases outward.
- An electron jumps to another shell by **absorbing or releasing energy equal to the difference** between the two levels. Each shell holds only a fixed number of electrons.
- Even Bohr's model has limits: the **modern (quantum mechanical) model** says electrons exist as **electron clouds**, and only the regions where they are most likely to be can be predicted.

| Model | Scientist | Main idea |
|---|---|---|
| **Indivisible atom** | Dalton (1808) | Atom is the smallest unit, cannot be divided |

| Model | Scientist | Main idea |
|---|---|---|
| **Plum pudding** | Thomson (1897) | Electrons embedded in a positive sphere |
| **Nuclear / planetary** | Rutherford (1911) | Small dense positive nucleus; electrons revolve around it |
| **Energy levels** | Bohr (1913) | Electrons in fixed shells without losing energy |
| **Quantum mechanical** | Modern | Electron clouds (probability regions) |

## 8.3 Neutron and Subatomic Particles

- Puzzle: a helium atom has 2 protons but about **4 times** the mass of hydrogen (1 proton). Reason: an uncharged particle in the nucleus - the **neutron**, discovered by **James Chadwick (1932)**.
- Neutron mass is nearly equal to proton mass. Found in the nucleus of all atoms **except hydrogen** (ordinary H). Mass of an atom comes mainly from **protons + neutrons**; electron mass is negligible.
- Heavier atoms have more neutrons than protons (Fe: 26p, 30n; U: 92p, 146n). Neutrons reduce repulsion between protons, and the **nuclear force** binds the particles together.

| Particle | Symbol | Relative charge | Location | Discovered by |
|---|---|---|---|---|
| **Electron** | $e^-$ | -1 | Outside nucleus (shells) | J. J. Thomson |
| **Proton** | $p^+$ | +1 | Nucleus | Rutherford |
| **Neutron** | $n^0$ | 0 | Nucleus | Chadwick |

## 8.4 Symbols of Elements

- Dalton (1803) gave the first **pictorial symbols**. **Berzelius (1813)**: symbols from **Latin names** (alphabets). Today **IUPAC** approves names and symbols (about 118 known elements).
- **Rules:** first letter capital, second (if any) small (Al, Co - not AL, CO). Often first one or two letters (H, Al); sometimes first letter + another letter (Cl, Zn); some from Latin/Greek/German names.

| Element | Symbol | Origin | Element | Symbol | Origin |
|---|---|---|---|---|---|
| Sodium | Na | Natrium | Silver | Ag | Argentum |
| Potassium | K | Kalium | Gold | Au | Aurum |
| Iron | Fe | Ferrum | Lead | Pb | Plumbum |
| Copper | Cu | Cuprum | Mercury | Hg | Hydrargyros (Greek) |
| Tungsten | W | Wolfram (German) | Others | N, O, S, Cl, Mg, Zn, Ca, Si | English names |

## 8.5 - 8.6 Atomic Number and Mass Number

- **Atomic number (Z)** = number of protons in the nucleus = number of electrons in a neutral atom. It **identifies the element** and decides its chemical behaviour.
- **Mass number (A)** = number of protons + number of neutrons. Protons and neutrons are together called **nucleons**. Electron mass is ignored.

**A = p + n | Number of neutrons = A - Z | Electrons = protons = Z (neutral atom)**
Notation: mass number A is written top-left and atomic number Z bottom-left of the symbol. Eg. carbon: A = 12, Z = 6 (written as $^{12}_{6}C$), so 6 protons, 6 neutrons, 6 electrons.

| Element | Protons | Neutrons | Mass number |
|---|---|---|---|
| **Hydrogen** | 1 | 0 | 1 |
| **Helium** | 2 | 2 | 4 |
| **Lithium** | 3 | 4 | 7 |

## 8.7 Electronic Configuration (Distribution of Electrons)

- **Bohr-Bury rules:** (1) maximum electrons in a shell = $\mathbf{2n^2}$ (K = 2, L = 8, M = 18); (2) the **outermost shell** can hold a maximum of **8** electrons; (3) shells fill in order **K, L, M, N...** - an outer shell fills only after the inner one is

complete.

- The distribution of electrons in shells is called **electronic configuration**.

| Element | Symbol | Z | Neutrons | Configuration (K, L, M) | Valence $e^-$ | Valency |
|---|---|---|---|---|---|---|
| **Hydrogen** | H | 1 | 0 | 1 | 1 | 1 |
| **Helium** | He | 2 | 2 | 2 | 2 | 0 |
| **Lithium** | Li | 3 | 4 | 2, 1 | 1 | 1 |
| **Beryllium** | Be | 4 | 5 | 2, 2 | 2 | 2 |
| **Boron** | B | 5 | 6 | 2, 3 | 3 | 3 |
| **Carbon** | C | 6 | 6 | 2, 4 | 4 | 4 |
| **Nitrogen** | N | 7 | 7 | 2, 5 | 5 | 3 |
| **Oxygen** | O | 8 | 8 | 2, 6 | 6 | 2 |
| **Fluorine** | F | 9 | 10 | 2, 7 | 7 | 1 |
| **Neon** | Ne | 10 | 10 | 2, 8 | 8 | 0 |
| **Sodium** | Na | 11 | 12 | 2, 8, 1 | 1 | 1 |
| **Magnesium** | Mg | 12 | 12 | 2, 8, 2 | 2 | 2 |
| **Aluminium** | Al | 13 | 14 | 2, 8, 3 | 3 | 3 |
| **Silicon** | Si | 14 | 14 | 2, 8, 4 | 4 | 4 |
| **Phosphorus** | P | 15 | 16 | 2, 8, 5 | 5 | 3 |
| **Sulfur** | S | 16 | 16 | 2, 8, 6 | 6 | 2 |
| **Chlorine** | Cl | 17 | 18 | 2, 8, 7 | 7 | 1 |
| **Argon** | Ar | 18 | 22 | 2, 8, 8 | 8 | 0 |

## 8.8 Valency (Combining Capacity)

- **Combining capacity** = number of H or Cl atoms with which one atom combines ($H_2O$: O = 2; $NH_3$: N = 3; $MgCl_2$: Mg = 2).
- **Valence shell** = outermost shell; electrons in it = **valence electrons**. A shell with 8 electrons is an **octet**. Octet (or 2 electrons for helium) means **stable and largely unreactive**.
- **Valency** = number of electrons an atom **gains, loses or shares** to complete its octet.
- Fewer than 4 valence electrons: **loses** electrons (Na 2,8,1: valency 1). More than 4: **gains** (O 2,6: valency 2). Exactly 4: **shares** (C 2,4: valency 4).

## 8.9 Isotopes, Average Atomic Mass and Isobars

### *Isotopes*

- Atoms of the **same element** with the **same atomic number but different mass numbers** (different number of neutrons). Eg. hydrogen: protium $^{1}H$ (0 n, 99.98%), deuterium $^{2}H$ (1 n), tritium $^{3}H$ (2 n). Carbon: $^{12}C$, $^{13}C$, $^{14}C$ (6 protons each).
- **Chemical properties are the same** (same electrons and configuration); **physical properties differ** (eg. melting and boiling points).

| Isotope | Use |
|---|---|
| **Uranium-235** | Fuel in nuclear reactors to generate electricity |
| **Cobalt-60** | Radiation treatment of cancer |
| **Iodine-131** | Treatment of goitre and thyroid cancer |
| **Carbon-14** | Finding age of ancient fossils and artefacts (archaeology, geology) |

### *Average atomic mass*

- Atomic mass is measured in **unified atomic mass unit (u)**. Natural elements are mixtures of isotopes, so the **weighted average** (mass × % abundance) is used, not a simple average.

- **Chlorine:** $^{35}Cl$ (75%) and $^{37}Cl$ (25%): 35 × 75/100 + 37 × 25/100 = **35.5 u**. (A simple average would wrongly give 36 u.) No single atom has mass 35.5 u; it is the average over a large sample.

### *Isobars*

- Atoms of **different elements** with the **same mass number but different atomic numbers**. Eg. calcium (Z = 20), potassium (Z = 19) and argon (Z = 18), each with mass number 40.

**Isotopes:** same Z, different A | **Isobars:** same A, different Z (different elements)

## Solved Examples (like textbook questions)

- Atomic number 26, 56 nucleons: protons = electrons = **26**, neutrons = 56 - 26 = **30**.
- Nucleus has 20 protons, mass number 41: neutrons = 41 - 20 = **21**.
- Atom with 18 neutrons and atomic number 17: mass number = **35** (chlorine, configuration 2, 8, 7, valency 1).
- Mass number 23, 11 protons: **sodium** (soft metal, reacts vigorously with water), neutrons = **12**.
- Magnesium (A = 24, Z = 12): 12 protons, 12 electrons, 12 neutrons; configuration **2, 8, 2**; valency 2.
- Bromine isotopes $^{79}Br$ (49.7%) and $^{81}Br$ (50.3%): average atomic mass = 79 × 0.497 + 81 × 0.503 = **about 80 u**.

## Remember for Exams

- Order of discovery: electron (Thomson) → nucleus and proton (Rutherford) → neutron (Chadwick). Models: Dalton → Thomson → Rutherford → Bohr → Quantum mechanical.
- Rutherford's model: nucleus is positive, dense, tiny; atom mostly empty. Failed on **stability**. Bohr fixed this with **stationary states**.
- Atomic number fixes the element; isotopes of an element differ only in neutrons.
- Valency from configuration: use number of valence electrons (if 4 or fewer) or 8 minus valence electrons (if more than 4). Noble gases (He, Ne, Ar) have valency 0.
- Neutral atom: protons = electrons; overall charge zero.